Evaluations and Benchmarks in Context of Multimodal LLM

https://mllm2024.github.io/CVPR2025/

CVPR Nashville JUNE 11-15, 2025

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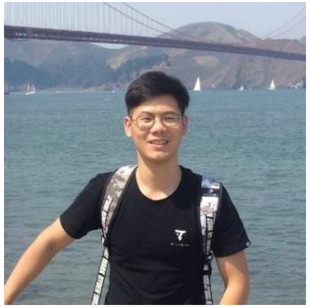

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Video Capability Evaluation

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Table of Content

Image and Video

- Understanding image and video data are two of the most important things to help us navigate the physical world. They have strong connections in visual information but also a lot of distinctions in many aspects.

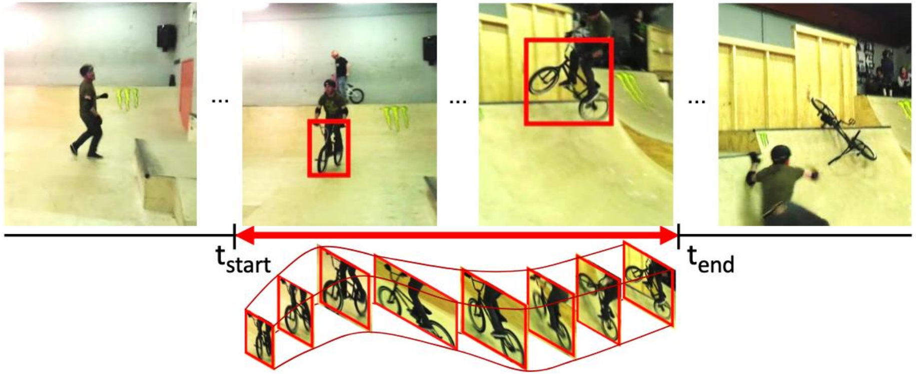


Image Understanding

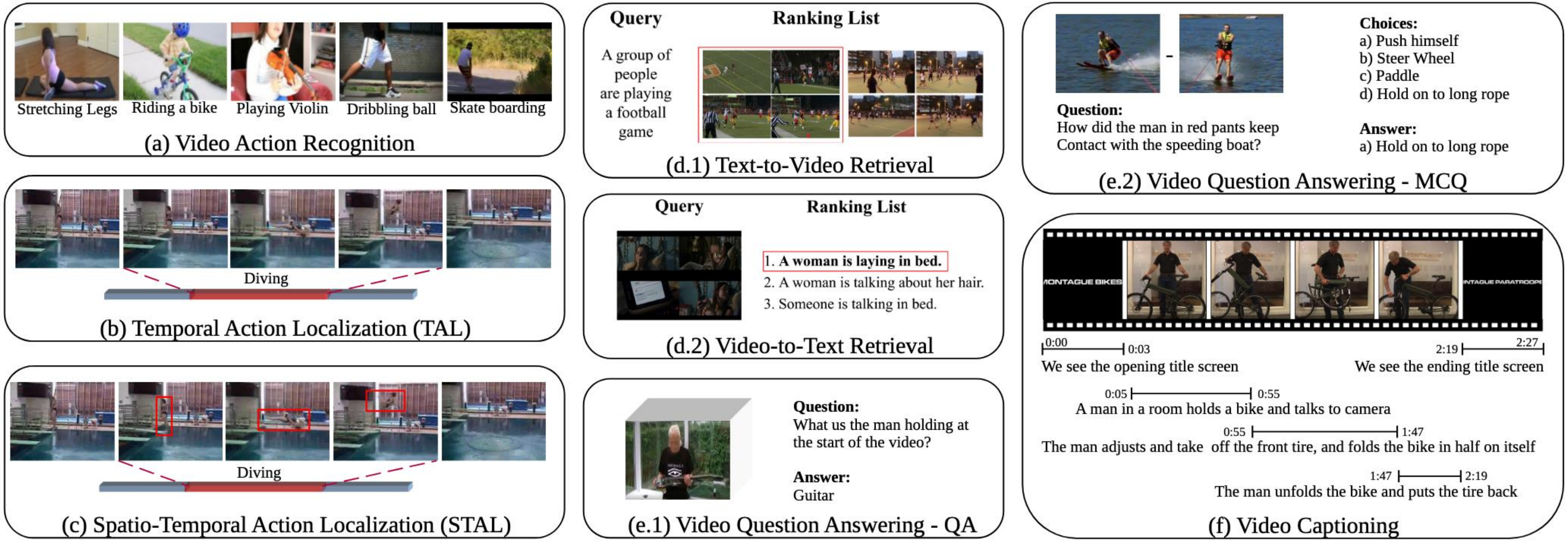
- Static visual information
- Objects, attributes, spatial, etc.
- Natively unimodal and lower complexity

Video Understanding

- Dynamic visual information
- Events, trajectories, temporal, etc.
- Natively multimodal and higher complexity

Traditional Video Understanding Tasks

- In the past decade, a large number of important video understanding tasks have emerged as separately studied lines of research, including video action recognition, temporal action localization, etc.



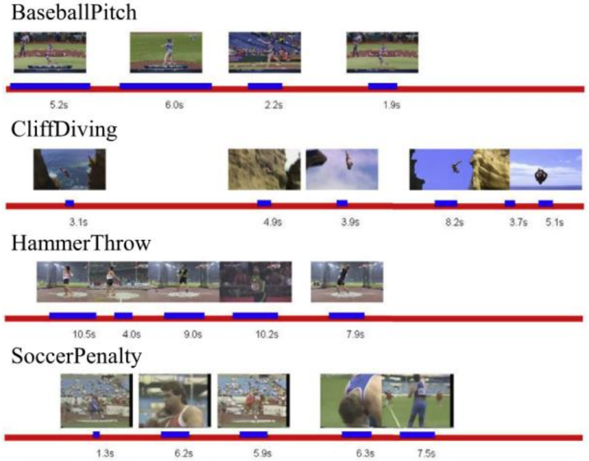
Video Capability Benchmarks

- A diverse range of downstream benchmarks are established to evaluate different aspects of video understanding capabilities, e.g., global content perception, local activity localization, visual scene recognition, and so on. However, it remains difficult to acquire and evaluate these capabilities **at the same time within a single model**.

Video Classification

- Global understanding
- Coarse-grained perception
- ……

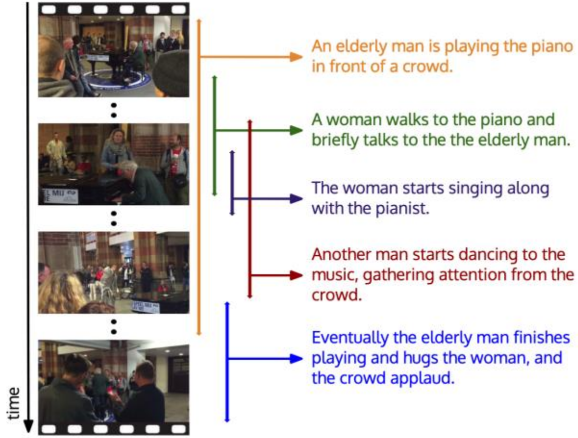
UCF-101, Kinetics-400 …



Action Detection

- Local understanding
- Fine-grained perception
- ……

THUMOS-14, ActivityNet v1.3 …



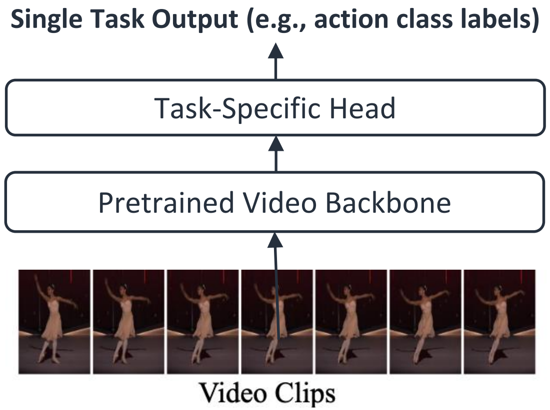
Video Captioning

- Activity Understanding
- Scene recognition
- ……

ActivityNet Captions, YouCook2, …

Specialist Video Models and VideoLLMs

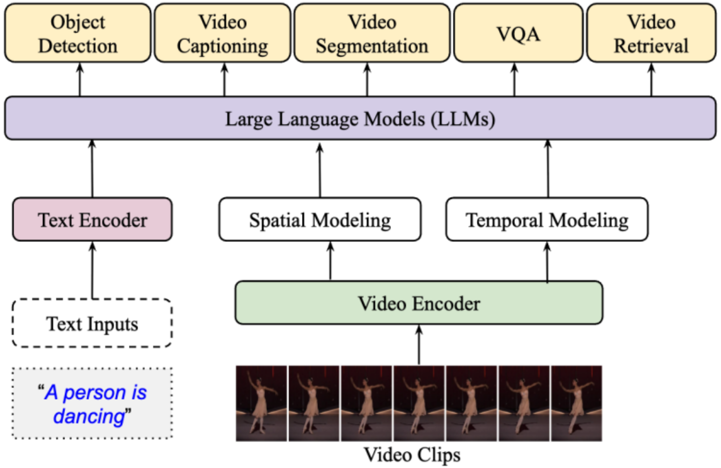
- Large Language Models (LLMs) have advanced rapidly in recent years. Meanwhile, single-task specialist video models are being supplanted by Video Large Language Models (VideoLLMs). These VideoLLMs fuse LLMs' strong text-processing power into one interface that can handle many kinds of video tasks.



Specialist Video Models

- Single-task capabilities
- Unimodal understanding
- Fixed output format

C3D, SlowFast, ...



Video Large Language Models

- Multi-task capabilities
- Multimodal understanding
- Flexible instruction following

Video-LLaVA, Video-ChatGPT, ...

Multimodal Benchmarks of VideoLLM

- **Capability**
  - Understanding
  - Reasoning
  - Generation
  - Application

Multimodal Benchmarks of VideoLLM

Capability

- **Understanding**
- Reasoning
- Generation
- Application

Understanding

- It refers to the ability to extract and integrate features from multimodal data to perform cross-modal analysis. This involves tasks such as interpreting visual representations, identifying key details, grasping semantic meaning, and responding accurately to related questions.
  - Visual Perception
  - Contextual Comprehension

Understanding

Visual Perception

- Visual perception capability is a foundational aspect of understanding benchmarks. It involves the ability to extract salient features and accurately recognize and interpret visual elements, such as multiple objects, text information, and complex emotional or implicit cues.
  - Fine-grained Perception
  - Coarse-grained Perception
  - Low-level Perception

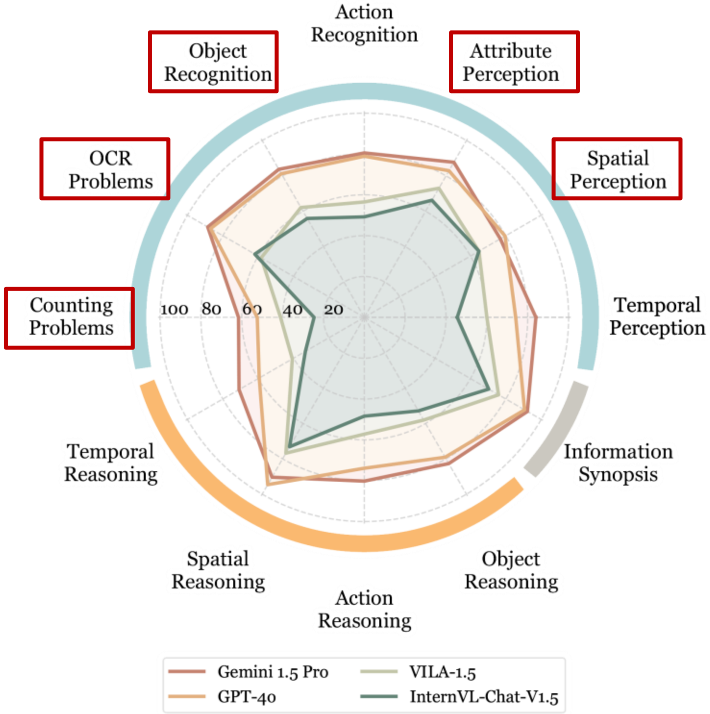
Understanding

Fine-grained Perception

Q: What color are the nails of a woman with white hair?
A: Pink or purple.

- OCR
- Object recognition
- Attribute recognition
- Position recognition
- Counting
- ……

Video-MME, Perception Test , MVBench, MMBench-Video , ……



Fine-grained Question Types in Video-MME

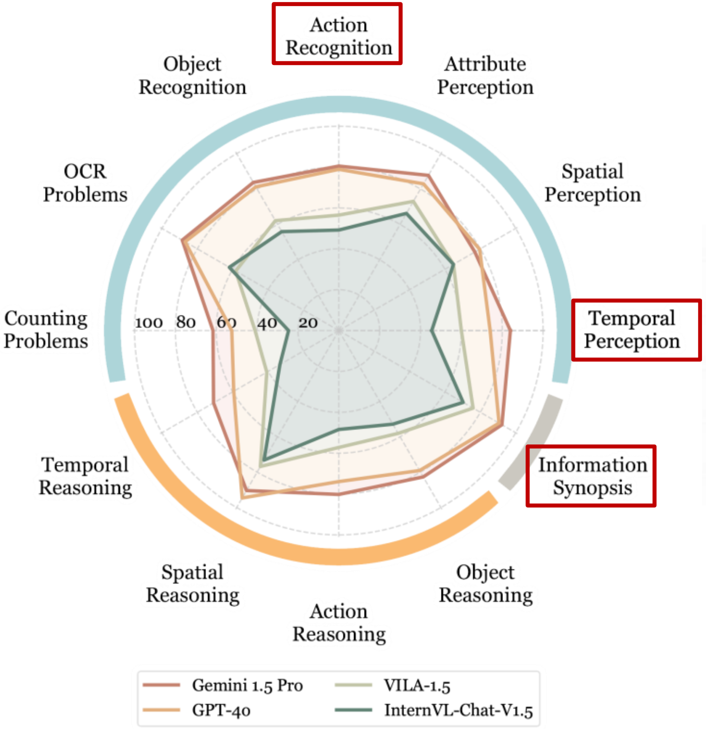
Understanding

Coarse-grained Perception

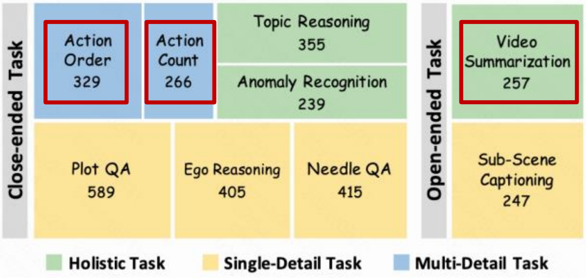
Q: What is this video about?
A: Tesla iOS App Integrates Shortcuts for Limitless Control

- Scene
- Event
- Topic
- Style
- Emotion
- ......

MMBench-Video, Video-MME



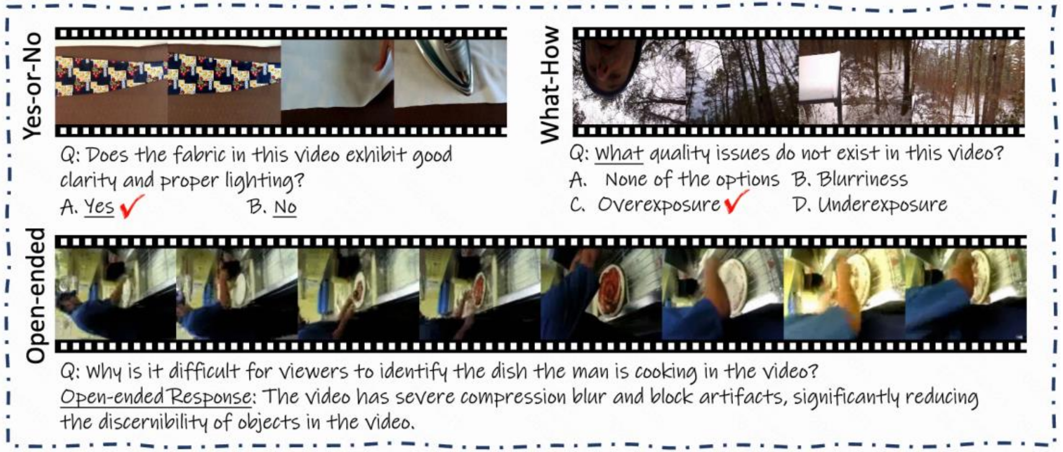
Coarse-grained Question Types in Video-MME

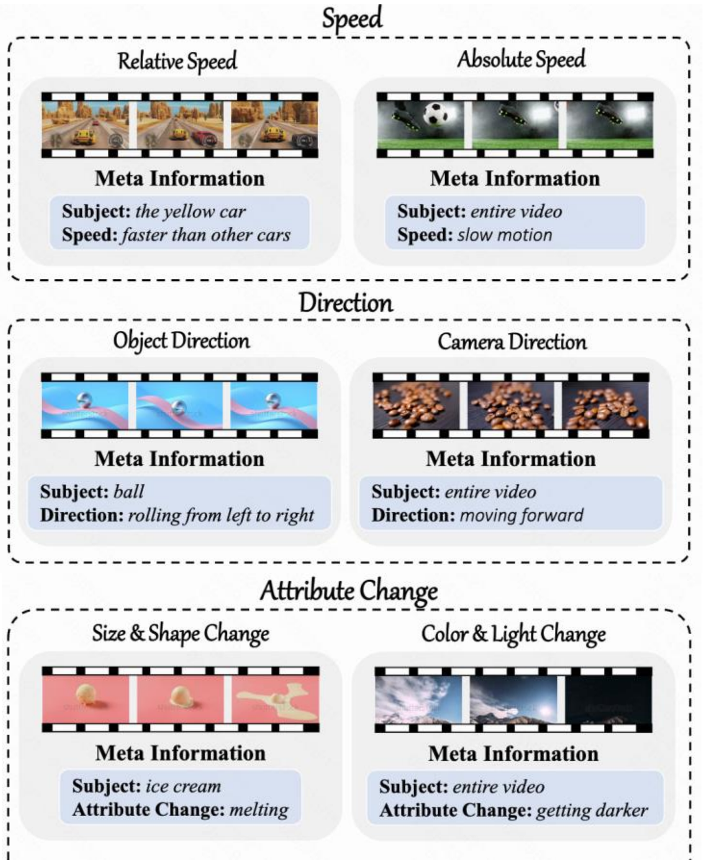


Coarse-grained Subset in MLVU

Understanding

Low-level Perception





- low-level degradation
- Aesthetic distortions
- Temporal Distortions
- ......

Q-Bench-Video, TempCompass, KVQ

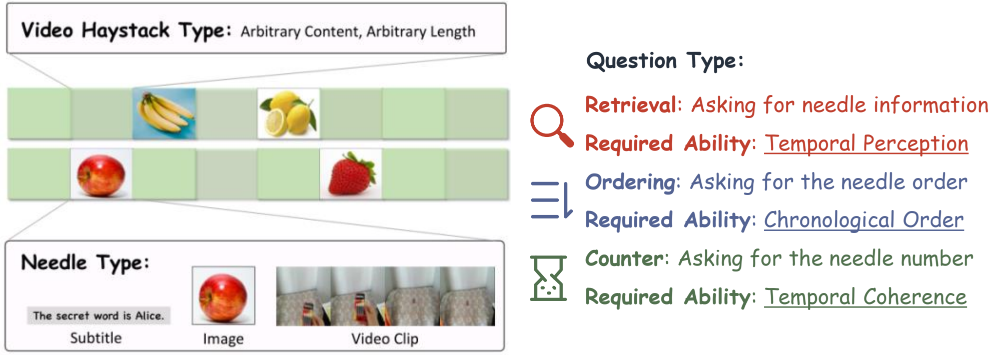
Understanding

Contextual Comprehension

- It refers to the ability of MLLMs to understand and interpret information that is influenced by the surrounding context. Based on the different input context formats, these benchmarks are grouped as follows:
  - Long Context Understanding
  - Interleaved Video-Language Understanding

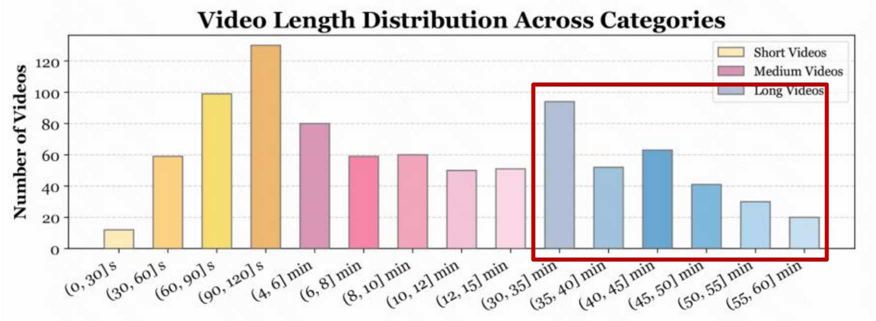
Understanding

Long Context Understanding

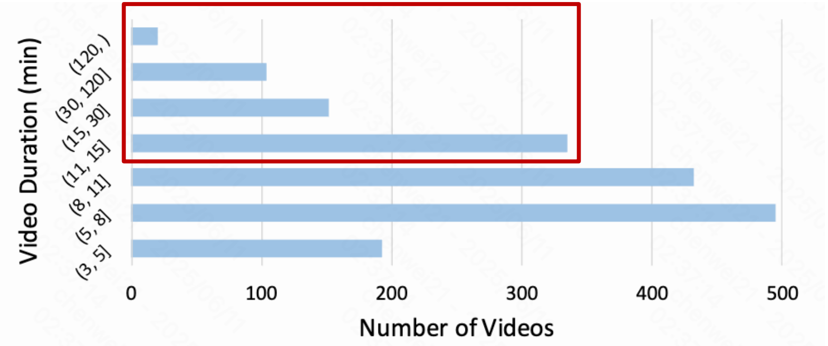


- Needle In A Haystack
- Event Detection
- ......

LongVideoBench, MLVU, Video-MME-M/L, VideoNIAH, EgoSchema, MMBench



Video-MME



MLVU

Understanding

Interleaved Video-Language Understanding

Question: At the beginning of the video, a woman with a headband tied to her head, wearing a red top, carrying a black backpack, when the woman comes down from a hill with tall rocks, what changes occur to her backpack?

Options:
- A. There is a dark red jacket hanging on her black backpack
- B. Nothing changed
- C. There is a white jacket hanging on her black backpack
- D. There is a dark blue jacket hanging on her black backpack

- Referred QA
- Video-Text
- Video-Audio
- Video-Audio-Text
- …

LongVideoBench, Video-MME, VideoMMMU, MMVU…

Benchmark	**Video frames**	**Subtitles / Text track**	**Raw audio**
LongVideoBench	✓	✓ (all videos come with human subtitles)	✗ (audio not provided)
Video-MME	✓	✓ (subtitle text)	✓ (original audio track released)
Video-MMMU	✓	✓ (transcribed audio used as text)	✓ (lecture narration)
MMVU	✓	✓ (“Text + Video” category in dataset table)	✗ (no audio requirement stated)

Understanding

Benchmark	Low-level	Fine-grained	Coarse-grained	Long-context	Interleaved V-L	Key focus
TempCompass	✓					Motion-tempo perception
Q-Bench-Video	✓					Video-quality distortions
VideoMME		✓	✓	✓	✓	Full-spectrum QA
MLVU			✓	✓	✓	Multi-task long-video
LongVideoBench				✓	✓	Hour-long interleaved QA
Video-MMMU				✓	✓	Lecture knowledge QA
MMVU				✓	✓	Expert-level reasoning

Multimodal Benchmarks of VideoLLM

- **Capability**
  - Understanding
  - **Reasoning**
  - Generation
  - Application

Reasoning

Understanding is necessary but not sufficient. Real-world queries demand causal, temporal, and commonsense inference.

Reasoning

Reasoning: output intermediate thinking steps first, then predict the final answer.

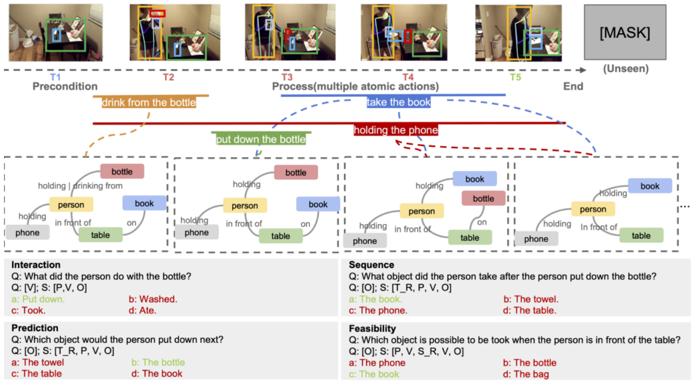
- Many tasks in video require reasoning:
  - **Long span**: evidence spread over hundreds of frames
  - **Concurrent events**: many objects/actions at once
  - **Occlusion**: key steps hidden or off-screen, must infer
  - **Multi-step goals**: must summarise, plan, or predict next steps

Reasoning

Reasoning Methods

Two types reasoning methods:

1. Base on Spatio-Temporal Scene Graph (STSG)



STEP (CVPR 25)
(2.5 + 1)D (AAAI 22)
…

2. End-to-end Reasoning (New trends)

Video-R1
VideoChat-R1
…

Reasoning

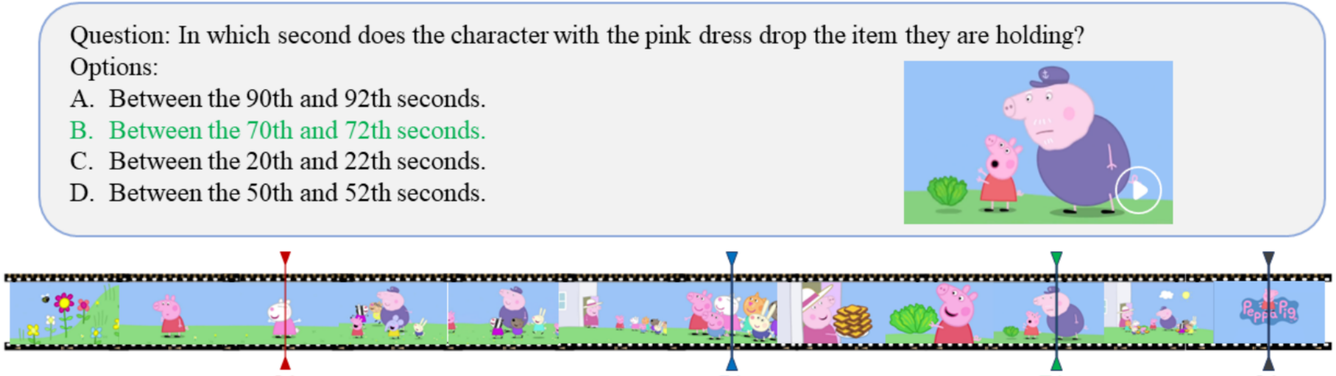
Reasoning Methods

Based on evaluation mode:

Mode	What is fixed?	Typical metric	When it's used
Objective (automatic)	A canonical **answer token** (label, span, time-window, count…) for every question	Accuracy / F1, IoU, BLEU, etc.	Classical VQA-style benchmarks, long-horizon retrieval, physics & counting tasks
Subjective (human / LLM judgment)	A **reasoning trace specification** (e.g. step-wise CoT, key-frame evidence) but not an exact wording	Human vote or GPT-4 "faithfulness"/"helpfulness" score	Benchmarks that require models to show their work or justify answers

Reasoning

Objective evaluation: closed questions solvable from explicit prompts or isolated visual cues.



(from VideoVista)

Advantages

Fully automated, fast scoring

Single ground-truth label → easy comparison

Accurately gauges basic perception & grounding

Scales to very large datasets

Limitations

Chain-of-thought ignored

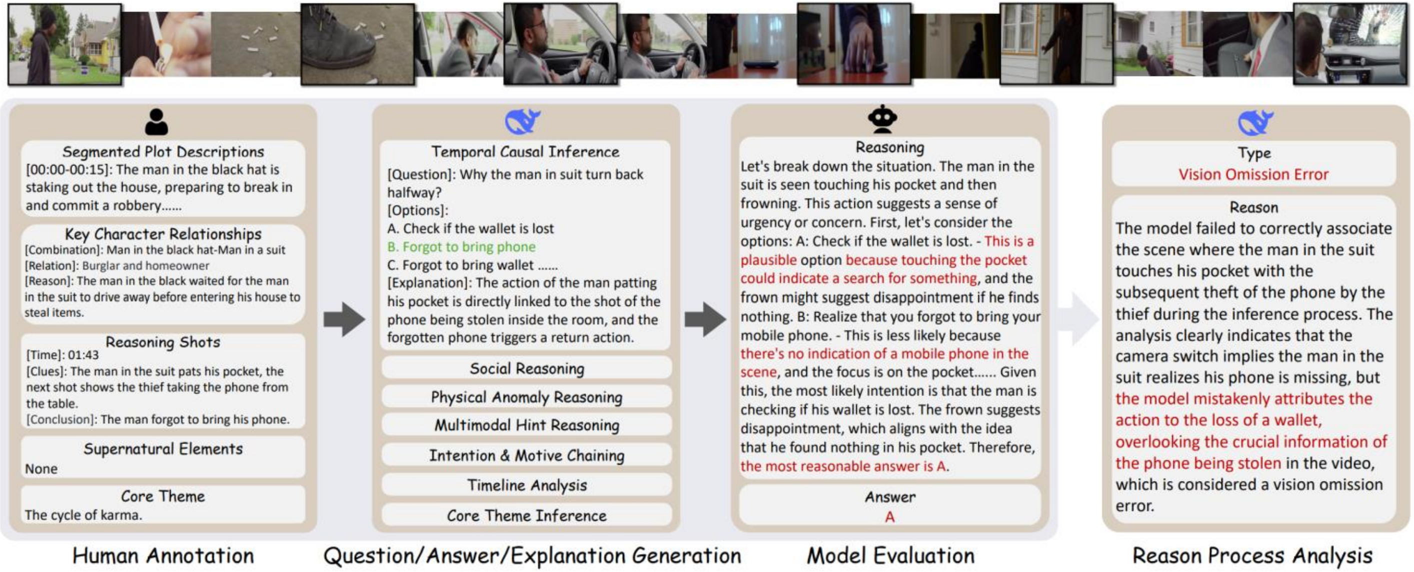
Guessing / dataset shortcuts still score high

Weak on multi-step, long-range reasoning

Little diagnostic insight when answers are wrong

Reasoning

Subjective evaluation: each question comes with lots of comprehensive ground-truth reasoning trace; humans or an LLM judge how closely the model's predicted reasoning matches that trace and assign a subjective score.



(from Video-Holmes)

Objective vs. Subjective Video-Reasoning Evaluation

	Objective (answer-only)	Subjective (answer + reasoning trace)
What's judged	Single ground-truth label	Alignment between model's chain-of-thought and a gold reasoning trace
Strengths	• Fully automated → fast, cheap, large-scale • Clear, reproducible metrics	• Mirrors real human reasoning • Reveals how and why the model reaches its answer • Better for diagnosing hallucinations
Weaknesses	• Can reward lucky guesses / shortcuts • No insight into internal logic • Poor coverage of multi-step, long-range tasks	• High annotation and review cost • Slower, harder to standardise • Subjective scores may vary across raters / LLM assessors

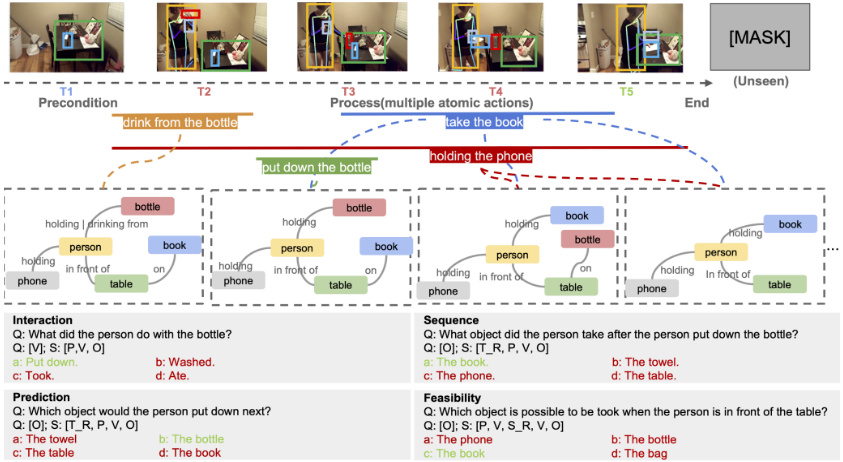
Reasoning

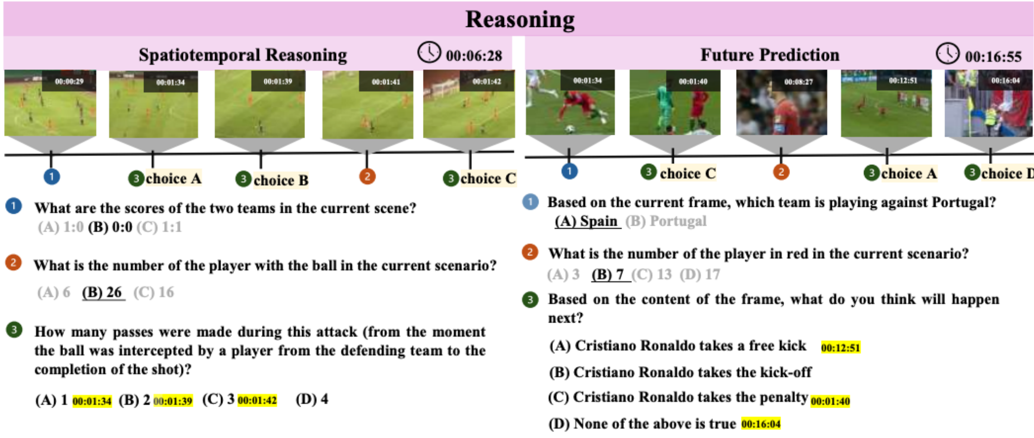
- Benchmarks can also be grouped by the target reasoning skill. It goes beyond basic understanding, encompassing the ability to perform complex inferences and draw logical conclusions across modalities. This includes tasks that require models to process and manipulate information, enabling them to solve problems and make decisions based on cross-modal data.
    - Relational Reasoning
    - Multi-step Reasoning
    - Knowledge-based Reasoning

Reasoning

Relational Reasoning

- Relational reasoning refers to the ability of MLLMs to recognize, manipulate, and reason about relationships between different entities or concepts.





Visual Spatial Relation

- Situated Reasoning
- Visual Relation Detection
- ……

STAR, VideoVista ...

Temporal Relation

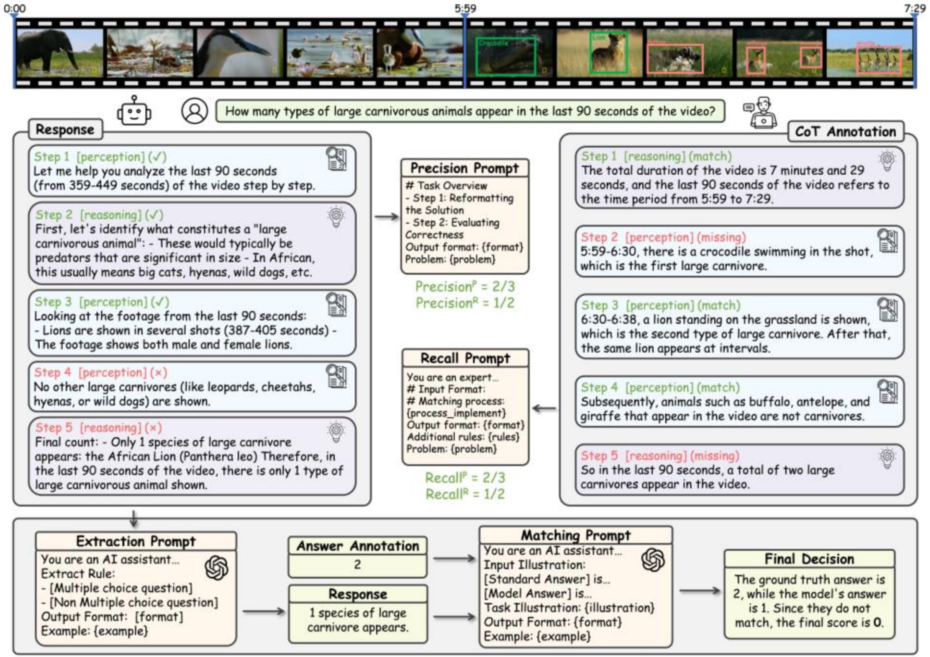
- Video QA
- Temporal grounding
- ……

ViLMA, ReXTime, RTV-Bench ...

Reasoning

Multi-Step Reasoning

- Multi-step Reasoning is crucial for complex cognitive tasks that necessitate navigating through a series of interconnected logical steps, e.g., chain of thoughts (CoT, which decomposes complex tasks into simpler, manageable subtasks).



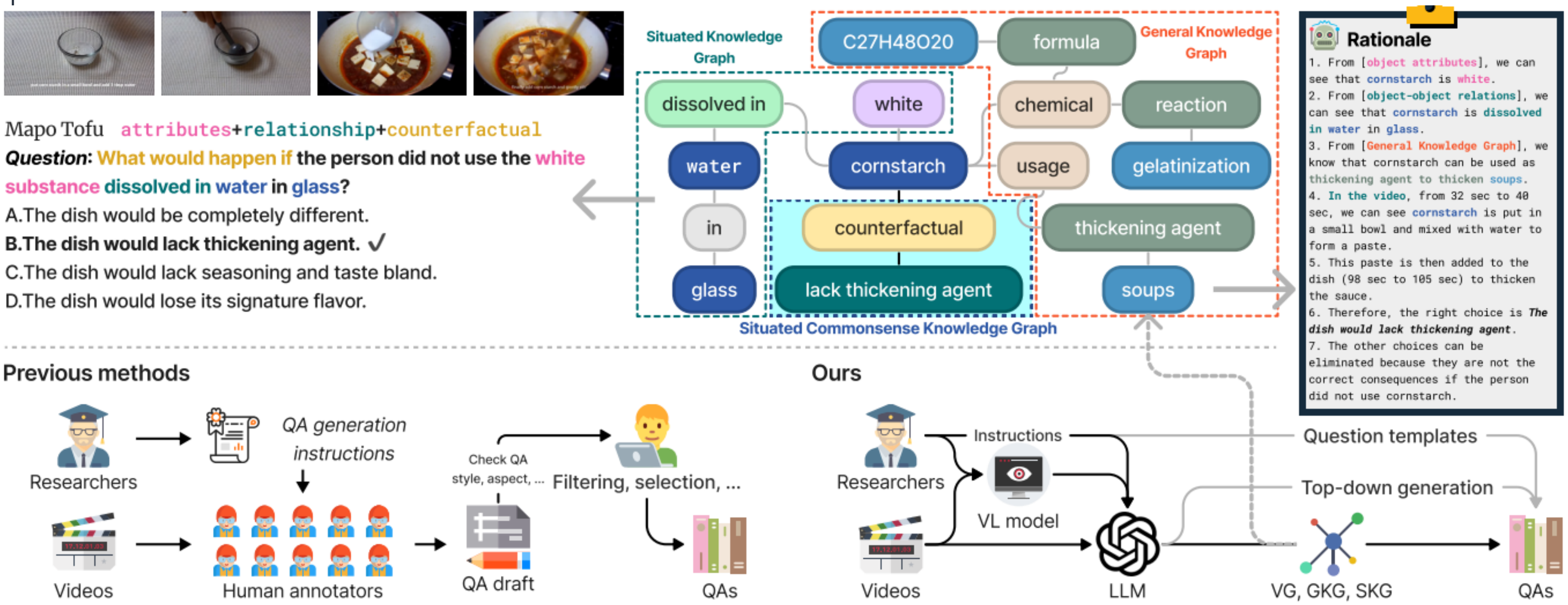
Chain-of-Thoughts

VideoCoT, VCR-Bench ...

Reasoning

Knowledge-based Reasoning

- Knowledge-based reasoning is the process of drawing conclusions or making decisions by leveraging structured information from an external knowledge graph, often incorporating commonsense knowledge answer complex queries.



Knowledge-based Reasoning

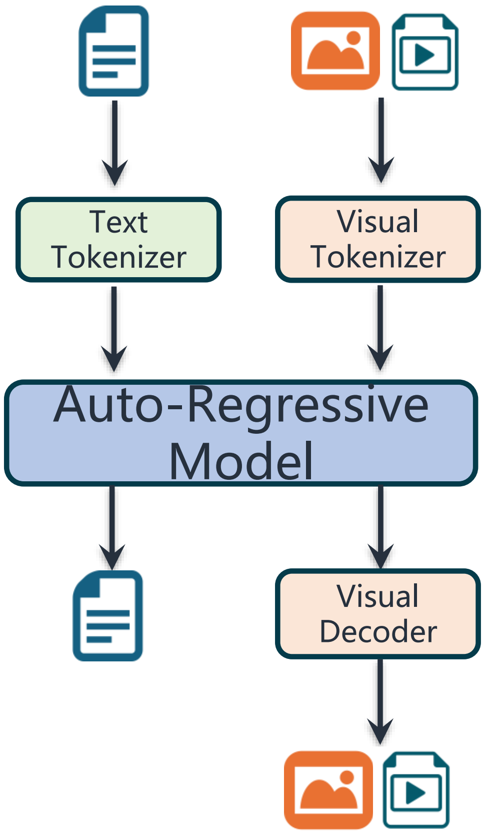
SOK-Bench ...

Multimodal Benchmarks of VideoLLM

- **Capability**
  - Understanding
  - Reasoning
  - **Generation**
  - Application

Generation

Current MLLM Frameworks that Supports Video Generation



Discrete Tokenizer (e.g.,VQ-VAE)

- VILA-U: a Unified Foundation Model Integrating Visual Understanding and Generation [ICLR 25]
- World Model on Million-Length Video and Language with Blockwise RingAttention [ICLR 25]

Continue Tokenizer (e.g., Diffusion Loss)

- Autoregressive Video Generation without Vector Quantization [ICLR 25]

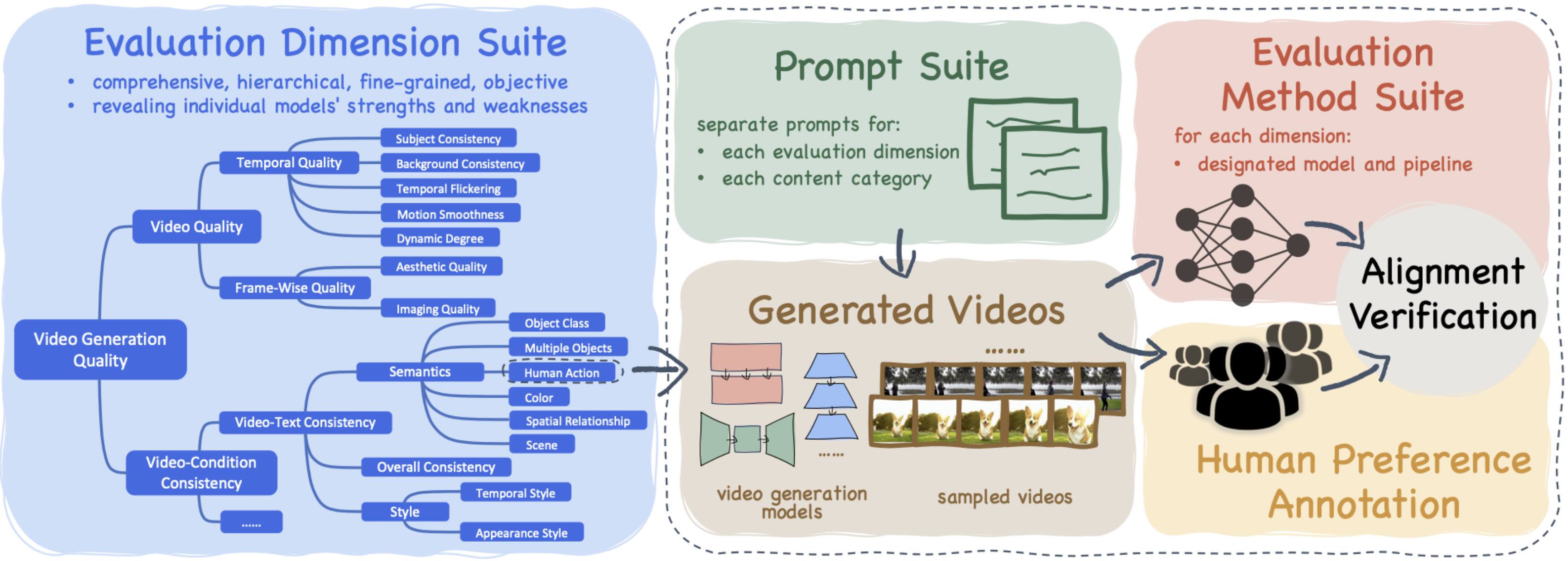
Token condition + Diffusion decoder

- NExT-GPT: Any-to-Any Multimodal LLM [ICML 24]
- WeGen: A Unified Model for Interactive Multimodal Generation as We Chat [CVPR 25]

Generation

Unified Models' Video Generation Benchmarks

Vbench

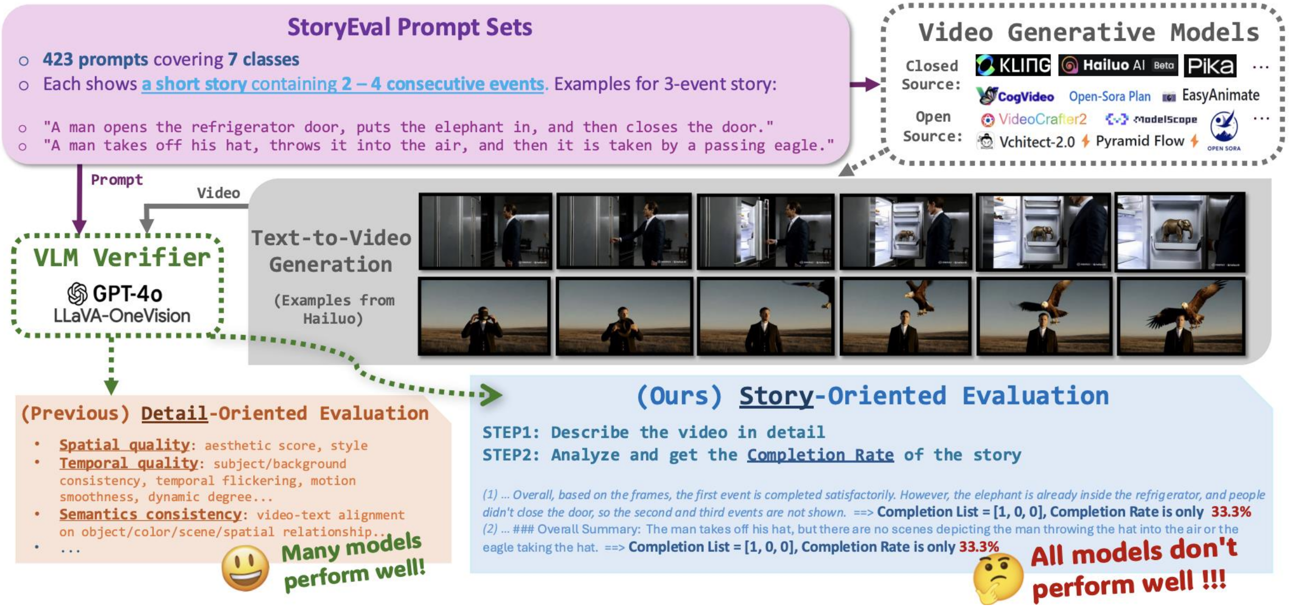


Advantages: fine-grained evaluation, multi-dimensional combination, combining automatic and manual evaluation.

Generation

Unified Models' Video Generation Benchmarks

StoryEval

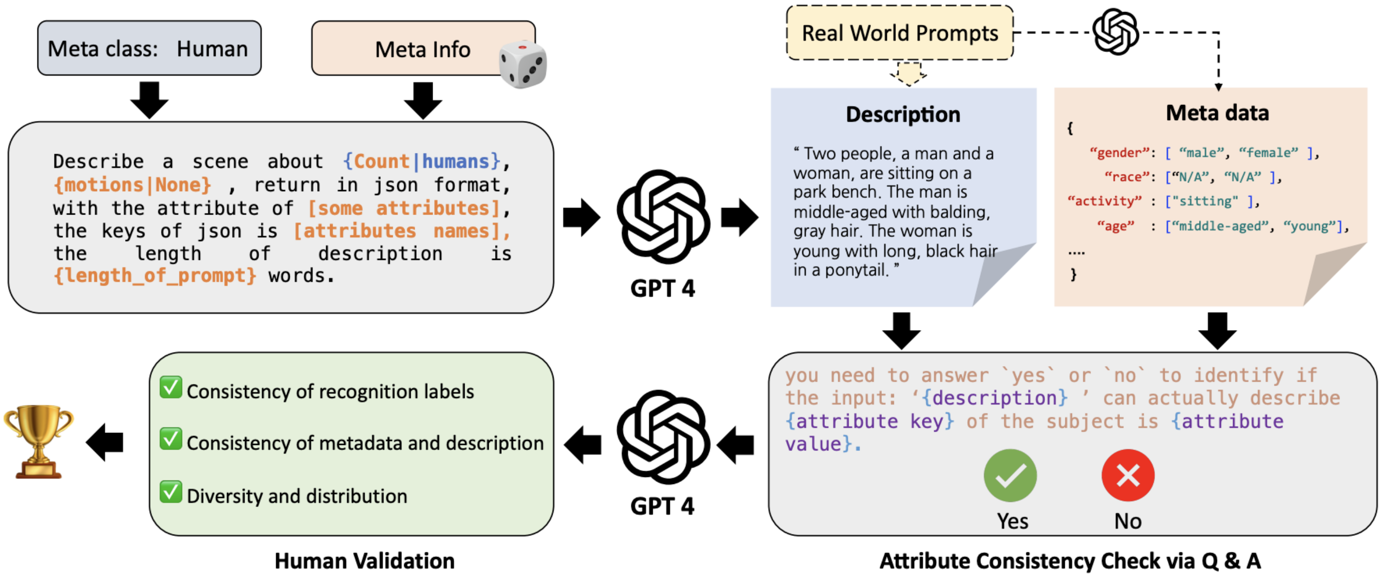


Advantages: Evaluate the "storytelling/world simulation" capabilities of video generation, focusing on event coherence and semantic consistency. Applicable to multi-event, complex scene generation evaluation.

Generation

Unified Models' Video Generation Benchmarks

EvalCrafter

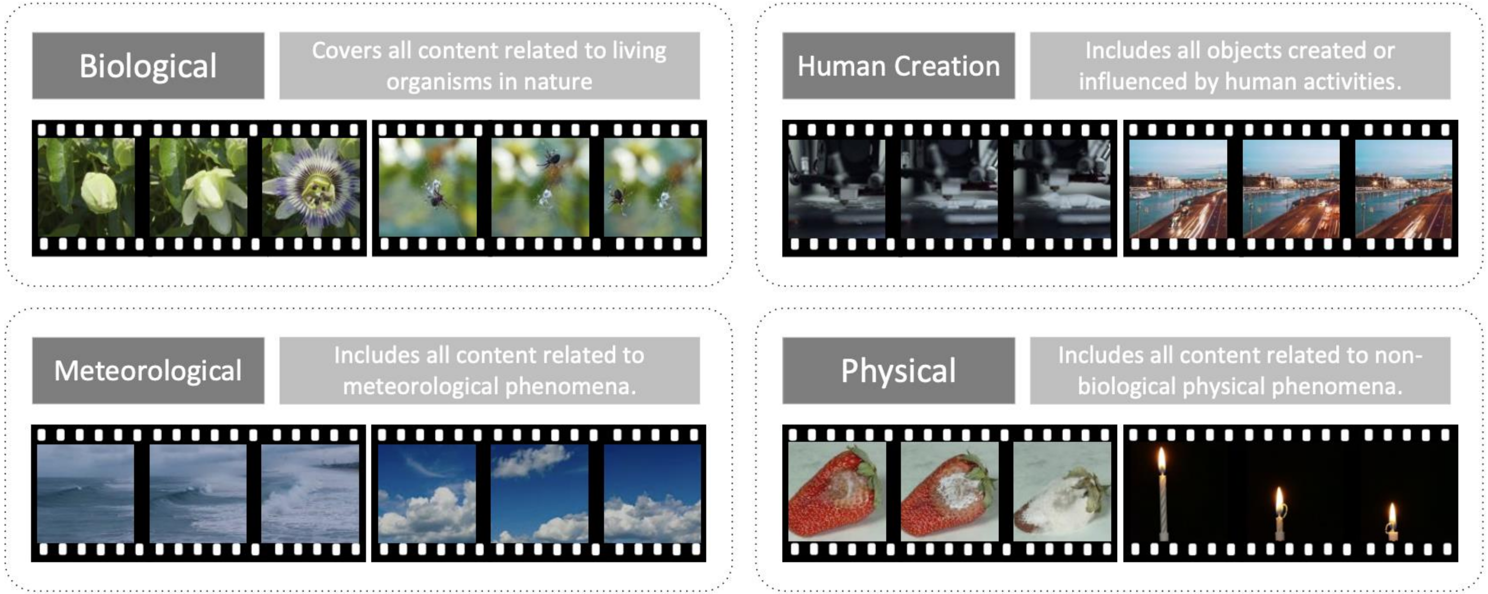


Advantages: Focus on core model capabilities such as semantic understanding, cross-modal alignment, and controllability.

Generation

Unified Models' Video Generation Benchmarks

ChronoMagic-Bench



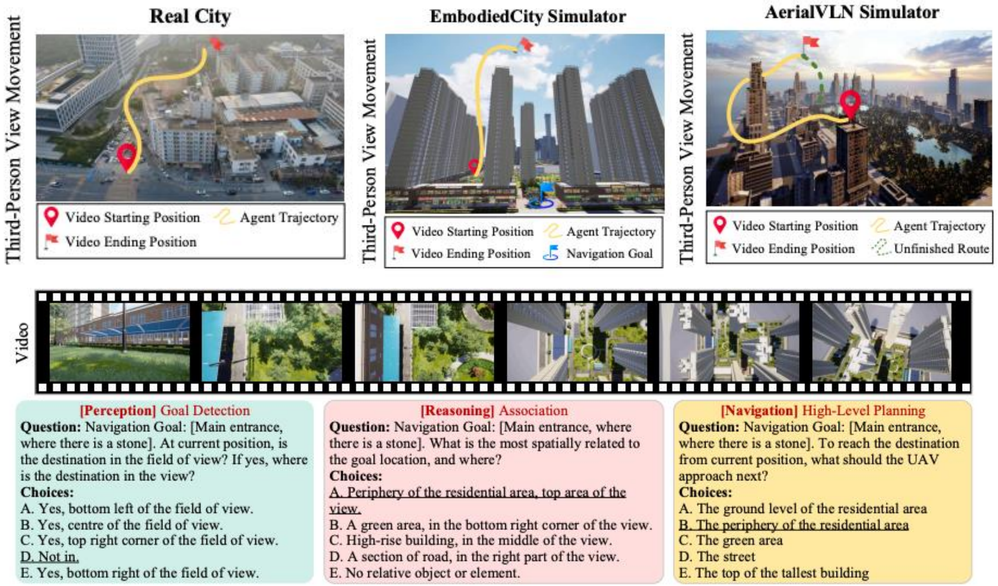
Advantages: Focusing on the performance of video generation models in terms of temporal changes, physical processes, and temporal coherence, it is particularly suitable for testing the model's ability to capture changes in the real physical world.

Multimodal Benchmarks of VideoLLM

- **Capability**
  - Understanding
  - Reasoning
  - Generation
  - **Application**

Application

Embodied AI Application



UrbanVideo-Bench, EmbodiedBench ...

Agent-based Application

T.	Question	Examples
Caption	Detailed Description	Q: *Please provide a detailed description of what occurs throughout these sequential GUI images.* A: The video shows a user taking the 16 Personalities test on a Windows desktop using the Edge browser...
	Summarized Caption	Q: *Write a clear description of the video, make sure the key features are well covered.* A: Creating a new IT team in Todoist by selecting industry, job function, role, team size, and inviting members.
Static	Layout, Icon Retrieval	Q: *What related searches are suggested on the right side of the Bing results for 'emnlp 2024'?* A: The suggested related searches shown include 'emnlp 2024 miami', 'eacl 2024 call for papers'...
	Textual Retrieval	Q: *What is the estimated time to complete the content for Week 2 of the course?* A: The estimated time to complete the content for Week 2 of the course is 1 hour...
	Interrelations in GUI Content	Q: *What is the name of the browser and the tab where the user performs the product search?* A: The browser is Microsoft Edge, and the user performs the product search in the eBay tab.
Dynamic	Content Retrieval	Q: *What specific action does the user take after turning their head to the left to view the left side of the page?* A: After turning their head to the left to view the left side of the page, the user performs...
	Prediction	Q: *Given the mouse is over 'Add NeurIPS 2024 DB Track Submission,' what's the likely next step?* A: It would be to click on the 'Add NeurIPS 2024 Datasets and Benchmarks Track Submission' button...
	Sequential Reasoning	Q: *Scrolls down from the 'Moon Gravity', which of the following cheats? A. Change Weather B. Skyfall ...* A: [[B]]

GUI-World, OSWORLD ...

Summary

Four capability pillars: Understanding, Reasoning, Generation, Application form the evaluation framework.

- **Understanding** – visual perception (low-, fine-, coarse-grained) and contextual comprehension (long-context & interleaved video-language).
- **Reasoning** – relational, multi-step (CoT) & knowledge-based inference; objective vs. subjective evaluation emerging.
- **Generation** – unified models judged by VBench, StoryEval, EvalCrafter & ChronoMagic-Bench for fidelity, coherence, controllability.
- **Application** – agent-based (GUI-World, OSWORLD) and embodied-AI (UrbanVideo-Bench, EmbodiedBench) tests close-loop task execution.

Thanks!

Any questions?